

Dynamic allocation + scheduling vs best measured allocation

Post-integration runtime | exhaustive allocation sweep | 7 settings shown

Workload	Worker budget	Model loop widths (L,R)	Best loop widths (L,R)	Model time (ms)	Best time (ms)	Extra time (ms)	Slower than best (%)
Compute loops + short tasks	4	(1, 1)	(3, 3)	1.3562	0.9079	+0.4483	+49.4%
Compute loops + short tasks	6	(1, 1)	(5, 5)	1.4922	0.6607	+0.8316	+125.9%
Unequal loops + short tasks	4	(1, 1)	(3, 1)	0.5206	0.2691	+0.2515	+93.4%
Unequal loops + short tasks	6	(1, 1)	(5, 1)	0.5285	0.2161	+0.3123	+144.5%
One long ordinary task	4	(1, 1)	(1, 1)	0.4359	0.3996	+0.0362	+9.1%
Three short ordinary tasks	4	(1, 1)	(3, 1)	0.2262	0.2124	+0.0139	+6.5%
Three short ordinary tasks	6	(1, 1)	(5, 1)	0.2262	0.2094	+0.0169	+8.1%

Gap = (model time / best measured time - 1) x 100. Times are medians of process medians.

Best time includes the best measured order and parent-worker limit; model time includes planning. Width 1 means serial.

Loop teams may run at different times. * = modal pair shown when choices vary; the CSV includes every observed choice.

Ordinary-first reference vs best measured allocation

Post-integration runtime | exhaustive allocation sweep | 7 settings shown

Workload	Worker budget	Model loop widths (L,R)	Best loop widths (L,R)	Model time (ms)	Best time (ms)	Extra time (ms)	Slower than best (%)
Compute loops + short tasks	4	(2, 1) *	(3, 3)	1.6196	0.9079	+0.7118	+78.4%
Compute loops + short tasks	6	(1, 1)	(5, 5)	1.4167	0.6607	+0.7560	+114.4%
Unequal loops + short tasks	4	(2, 1)	(3, 1)	0.6903	0.2691	+0.4212	+156.5%
Unequal loops + short tasks	6	(1, 1)	(5, 1)	0.6193	0.2161	+0.4032	+186.5%
One long ordinary task	4	(1, 1)	(1, 1)	0.6846	0.3996	+0.2850	+71.3%
Three short ordinary tasks	4	(1, 1)	(3, 1)	0.4670	0.2124	+0.2547	+119.9%
Three short ordinary tasks	6	(1, 1)	(5, 1)	0.3061	0.2094	+0.0967	+46.2%

Gap = (model time / best measured time - 1) x 100. Times are medians of process medians.

Best time includes the best measured order and parent-worker limit; model time includes planning. Width 1 means serial.

Loop teams may run at different times. * = modal pair shown when choices vary; the CSV includes every observed choice.